

$$P^{(3)}(H) := P^{(3)}(H|E_3) = \frac{P^{(0)}(HE_3|E_3E_1)}{P^{(0)}(E_3|E_3E_1)}$$

$$= P^{(0)}(H|E_3E_1)$$

...

$$P^{(n)}(H) := P^{(n-1)}(H|E_n)$$

$$= P^{(0)}(H|E_nE_3E_1)$$

$$= P(H|E_nE_3E_1)$$

$$3. P_n(H) = P'_n(H)$$

(data order invariant)

$$\text{here } P_0(H) := P(H)$$

$$\begin{aligned} P_1(H) &:= P(H|E_1) = \frac{P_0(H)P(E_1|H)}{Q(E_1; P_0)} \\ &= \frac{P(H)P(E_1|H)}{Q(E_1; P_0)} \end{aligned}$$

$$Q(E_1; P_0) = \sum_H P_0(H)P(E_1|H)$$

$$= \sum_H P(H)P(E_1|H) = P(E_1)$$

$$\therefore EE_3 = \{1, 2\}, EE_3 = \{2\}, EE_3 = \{2, 5, 6\}$$

$$EE_3 = \{2\}$$

$$\therefore P(E)P(E) = P(EE)$$

$$P(E_3)P(EE) = P(EE_3E_3)$$

$$\text{but } P(E)P(E_3) \neq P(EE_3)$$

$$P(E)P(E_3) \neq P(EE_3)$$

$$2. P^{(0)}(H) := P(H)$$

$$P^{(1)}(H) := P^{(0)}(H|E_1)$$

$$P^{(2)}(H) := P^{(1)}(H|E_2)$$

$$= \frac{P^{(0)}(HE_2)}{P^{(0)}(E_2)}$$

$$= \frac{P^{(0)}(HE_2|E_1)}{P^{(0)}(E_2|E_1)} \cdot \frac{P^{(0)}(E_1)}{P^{(0)}(E_1)}$$

$$= \frac{P^{(0)}(HE_2E_1)}{P^{(0)}(E_2E_1)}$$

$$= P^{(0)}(H|E_2E_1)$$

updating info sequentially

$$E_1, E_1 \text{ indep.}$$

$$E_2, E_2 \text{ indep.}$$

...

$$E_1, E_2, \dots, E_n \text{ indep.}$$

$$\Rightarrow P(E_1 \dots E_n) = P(E_1)P(E_2) \dots P(E_n)$$

$$\text{proof: } P(EE_3E_1)$$

$$= P(E_1E_2E_3)P(E_1) \quad (E_1, E_2, E_3 \text{ indep.})$$

$$= P(E_1 \dots E_{n-2})P(E_{n-1})P(E_n) \quad (E_1, E_2, \dots, E_n \text{ indep.})$$

...

$$= P(E_1E_2)P(E_3) \dots P(E_n)$$

$$= P(E_1)P(E_2) \dots P(E_n) \quad (E_1, E_2, E_3 \text{ indep.})$$

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$$\left. \begin{array}{l} E_1, E_2 \text{ indep.} \\ E_3, E_1E_2 \text{ indep.} \end{array} \right\} \Rightarrow E_1, E_2, E_3 \text{ indep.}$$

example: $\{1, 2, \dots, 8\}$, each probability $\frac{1}{8}$.

$$E_1 = \{1, 2, 3, 4\}, E_2 = \{1, 2, 5, 6\}, E_3 = \{2, 5, 6, 7\}$$

$$\therefore \sum_H P(H|EE_n) = 1$$

$$\therefore k_3 = \sum_H f_3(H)$$

$$\therefore P_n(H) = \frac{f_1(H)}{k_1} \quad (\text{from 3})$$

$$k_1 = \sum_H f_1(H)$$

$$\therefore f_1(H) = f_3(H)$$

$$\therefore k_1 = k_3 \therefore P(H|EE_n) = P_n(H)$$

□

$$5. P(H|H_1) = \frac{1}{2}, P(T|H_1) = \frac{1}{2},$$

$$P(H|H_2) = \frac{4}{5}, P(T|H_2) = \frac{1}{5},$$

$$P(H_1) = P(H_2) = \frac{1}{2},$$

flip conditionally indep given coin $H_i, i=1,2$.

if observe H_1, H_1, T , i.e., $E_1=H, E_2=H, E_3=T$

solve P_1, P_2, P_3 .

$$\text{solve: } P_1(H_1) = \frac{5}{13}, P_1(H_2) = \frac{8}{13}$$

$$P_2(H_1) = \frac{25}{89}, P_2(H_2) = \frac{64}{89}$$

$$P_3(H_1) = \frac{125}{253}, P_3(H_2) = \frac{128}{253}$$

$$P'_n(H) = \frac{P(H)P(E_1|H) \cdots P(E_n|H)}{Q(E_1; P'_1)Q(E_2; P'_1)Q(E_3; P'_2) \cdots Q(E_n; P'_n)}$$

$$= \frac{f_2(H)}{k_2}$$

$$\therefore \sum_H P_n(H) = 1, \sum_H P'_n(H) = 1.$$

$$\therefore k_1 = \sum_H f_1(H), k_2 = \sum_H f_2(H)$$

$$\therefore f_1(H) = f_2(H) \therefore k_1 = k_2$$

$$\therefore P_n(H) = P'_n(H) \quad \square$$

$$4. P(H|E_1 \cdots E_n) = P_n(H)$$

$$\text{if } P(E_1 \cdots E_n|H) = P(E_1|H) \cdots P(E_n|H)$$

$$\text{proof: } P(H|E_1 E_n) = \frac{P(H E_1 E_n)}{P(E_1 E_n)}$$

$$= \frac{P(H)P(E_1|H)}{P(E_1 E_n)}$$

$$= \frac{P(H)P(E_1|H)P(E_n|H) \cdots P(E_n|H)}{P(E_1 \cdots E_n)}$$

$$= \frac{f_3(H)}{k_3}$$

$$P_2(H) := \frac{P(H)P(E_2|H)}{Q(E_2; P_1)}$$

$$= \frac{P(H)P(E_1|H)P(E_2|H)}{Q(E_1; P_0)Q(E_2; P_1)}$$

$$Q(E_2; P) = \sum_H P(H)P(E_2|H)$$

$$(\neq P(E_2) = \sum_H P(H)P(E_2|H))$$

$$P_3(H) := \frac{P_2(H)P(E_3|H)}{Q(E_2; P_2)} = \frac{P(H)P(E_1|H)P(E_2|H)P(E_3|H)}{Q(E_1; P_0)Q(E_2; P_1)Q(E_3; P_2)}$$

$$Q(E_2; P_2) = \sum_H P_2(H)P(E_2|H) (\neq P(E_2))$$

$$P_n(H) := \frac{P_{n-1}(H)P(E_n|H)}{Q(E_{n-1}; P_{n-1})} = \frac{P(H)P(E_1|H) \cdots P(E_n|H)}{Q(E_1; P_0)Q(E_2; P_1) \cdots Q(E_{n-1}; P_{n-1})}$$

$$Q(E_n; P_n) = \sum_H P_n(H)P(E_n|H)$$

$P'_n(H)$ is obtained by E_1', E_2', E_n' .

$1', 2', \dots, n'$ is permutation of $1, 2, \dots, n$.

$$\text{proof: } P_n(H) = \frac{P(H)P(E_1|H)P(E_n|H)}{Q(E_1; P_0) \cdots Q(E_n; P_n)} = \frac{f(H)}{k_n}$$